Research Questions:

1. **What are the main air pollutants in Delhi, and what are their trends over time?**
2. **How do the concentrations of these pollutants vary, and what can we infer from their distributions?**
3. **What are the health and environmental impacts of these pollutants?**
4. **What strategies can be recommended to mitigate pollution levels in Delhi?**

1. Data Overview and Preparation:

The dataset contains AQI data from January 2023, with 561 entries and the following columns:

- **Date**: Timestamps (YYYY-MM-DD HH:MM format)
- **CO (Carbon Monoxide)**: Levels in µg/m³
- **NO2 (Nitrogen Dioxide)**: Levels in µg/m³
- **O3 (Ozone)**: Levels in µg/m³
- **PM2.5 (Particulate Matter 2.5)**: Levels in µg/m³
- **PM10 (Particulate Matter 10)**: Levels in µg/m³
- **NH3 (Ammonia)**: Levels in µg/m³

Data Loading:

The data was loaded using `pandas`, ensuring the date column was parsed as a datetime object.

2. Descriptive Statistics:

Descriptive statistics give us an overview of the pollutant concentrations throughout the month:

Pollutant	Min	Max	Mean	Std Dev	25%	50%	75%
CO	654	16,876	3,814	3,227	1,709	2,590	4,433
NO2	13.4	263.2	75.29	42.47	44.55	63.75	97.33
O3	0	164.5	30.14	39.98	0.07	11.80	47.21
PM2.5	60.1	1,310	358.26	227.36	204.45	301.17	416.65
PM10	69.1	1,499	420.99	271.29	240.9	340.9	482.57
NH3	0.63	267.5	26.43	36.56	8.23	14.82	26.35

- **CO** levels show a significant variation, with spikes reaching up to 16,876 µg/m³.
- **NO2** and **O3** concentrations are more moderate, but the distributions indicate possible extreme events.
- **PM2.5** and **PM10**, known as harmful particulate matter, also show concerning maximum values, with averages above safe limits.
- **NH3** is relatively low in comparison but still presents a notable distribution spread.

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3. Data Visualizations:

Pollutant Distributions:

The histogram plots of each pollutant show right-skewed distributions (see attached image), suggesting that while low or moderate levels are common, there are significant peaks in concentrations.

- **CO** has a wide range of values, with frequent spikes in high concentrations.
- **NO2** shows a similar pattern with occasional high values.
- **O3** generally remains at low levels, but there are some extreme cases.
- **PM2.5** and **PM10** consistently show high concentrations, indicating severe air quality issues related to particulate matter.
- **NH3** levels remain mostly low but are concerning when they spike.

Time Series Analysis:

The time series analysis over January (see the trends plot) indicates:

- **CO** has multiple significant spikes, suggesting high pollution events.
- **PM2.5** and **PM10** stay consistently high, with only slight fluctuations.
- **NO2** shows a similar trend to CO, with occasional peaks, indicating possible traffic or industrial activities as a source.

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4. Impact on Health and Environment:

- **PM2.5 and PM10**: Both particulate matters are highly harmful as they can penetrate deep into the lungs and bloodstream, causing respiratory issues, cardiovascular diseases, and even premature death. The consistently high levels observed are far above the safe limits recommended by the World Health Organization (WHO), indicating a severe public health risk.
- **CO**: High levels of carbon monoxide reduce the amount of oxygen that can be transported in the bloodstream to critical parts of the body. Acute exposure can lead to poisoning and long-term health issues.
- **NO2**: Elevated levels of nitrogen dioxide can irritate the airways in the human respiratory system and increase the risk of respiratory infections.
- **O3**: Ground-level ozone can cause or exacerbate respiratory problems, including asthma, bronchitis, and other lung diseases.
- **NH3**: Ammonia is usually associated with agricultural activity and waste, and though not as harmful at low levels, high exposure can cause irritation of the eyes, nose, and throat.

5. CODE

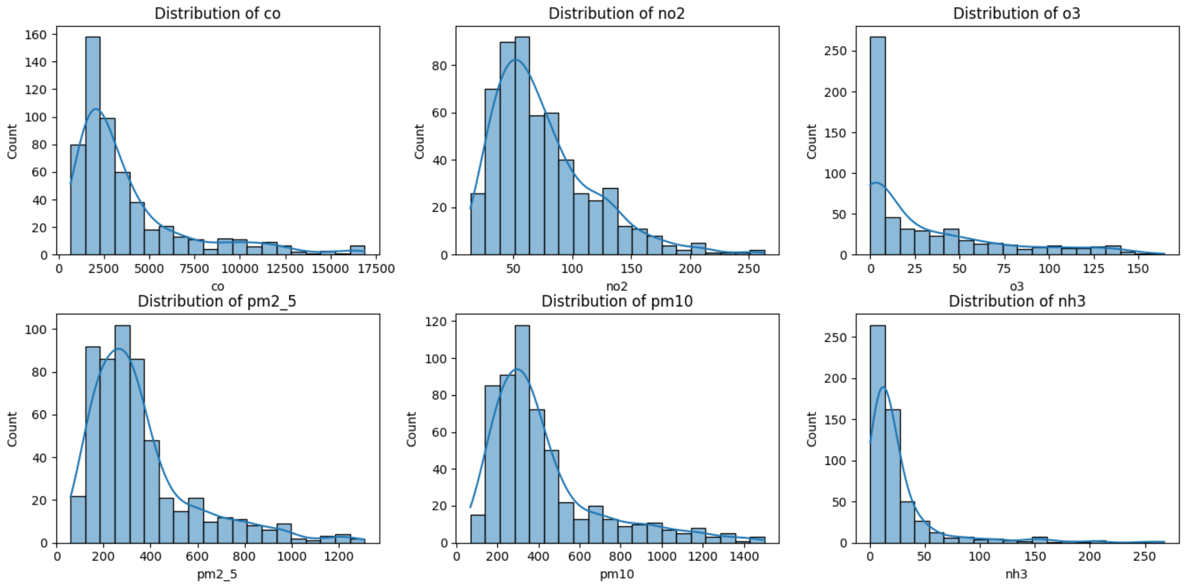
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from bokeh.plotting import figure, show, output_file
from bokeh.io import output_notebook
from bokeh.models import HoverTool

# Load the dataset
df = pd.read_csv('C:/Users/Admin/Desktop/delhiaqi.csv', parse_dates=['date'])

# Display basic information about the dataset
print(df.info())
print(df.describe())

# Descriptive statistics
print(df.describe())

# Calculate the correlation matrix
corr = df[['co', 'no2', 'o3', 'pm2_5', 'pm10', 'nh3']].corr()

# Create a heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Between Pollutants')
plt.show()

# Plot the distribution of each pollutant
plt.figure(figsize=(12, 8))
for i, column in enumerate(['co', 'no2', 'o3', 'pm2_5', 'pm10', 'nh3']):
    plt.subplot(2, 3, i+1)
    sns.histplot(df[column], bins=20, kde=True)
    plt.title(f'Distribution of {column}')
```

```python
    plt.title(f'Distribution of {column}')
plt.tight_layout()
plt.show()

# Activate output in the notebook
output_notebook()

# Create the Bokeh plot
p = figure(x_axis_type='datetime', title="Pollutants Over Time", width=800, height=400)
p.xaxis.axis_label = 'Date'
p.yaxis.axis_label = 'Concentration (µg/m³)'

# Add lines for pollutants with hover tool
pollutants = ['CO', 'NO2', 'O3', 'PM2_5', 'PM10', 'NH3']
colors = ['blue', 'green', 'orange', 'red', 'purple', 'brown']
for pollutant, color in zip(pollutants, colors):
    p.line(df['date'], df[pollutant], legend_label=pollutant, line_width=2, color=color)

p.add_tools(HoverTool())
p.legend.location = "top_left"

# Show the plot
show(p)
```

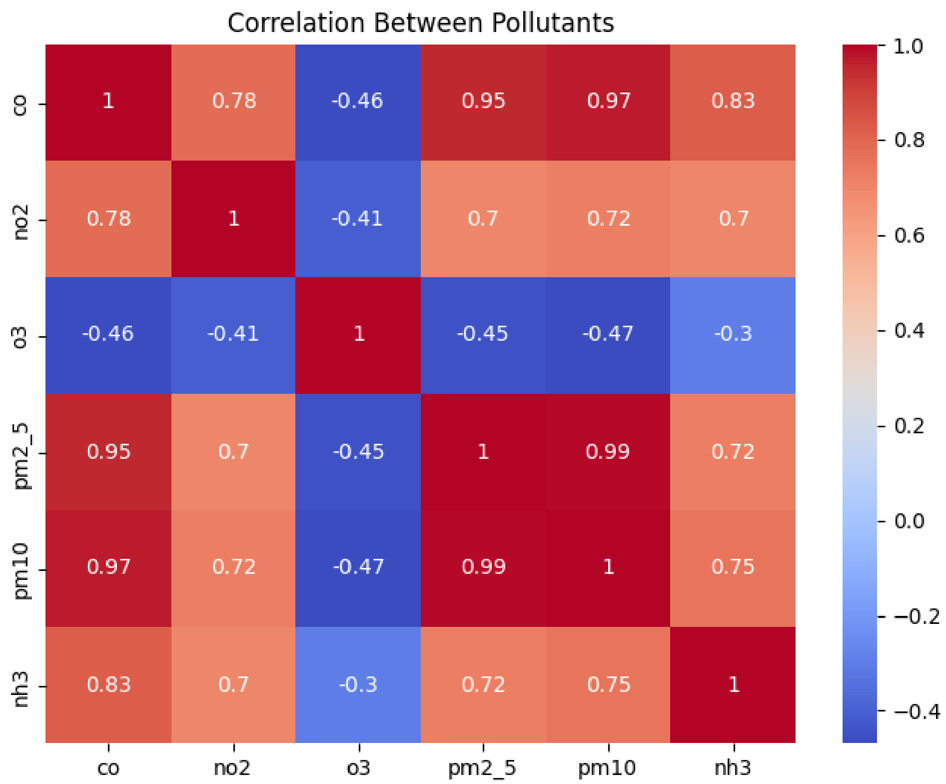
6. OUTPUT AND GRAPHS

```
Data columns (total 9 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   date    561 non-null    datetime64[ns]
 1   co      561 non-null    float64
 2   no      561 non-null    float64
 3   no2     561 non-null    float64
 4   o3      561 non-null    float64
 5   so2     561 non-null    float64
 6   pm2_5   561 non-null    float64
 7   pm10    561 non-null    float64
 8   nh3     561 non-null    float64
dtypes: datetime64[ns](1), float64(8)
memory usage: 39.6 KB
None
                      date            co  ...         pm10         nh3
count                  561    561.000000  ...   561.000000  561.000000
mean   2023-01-12 16:00:00   3814.942210  ...   420.988414   26.425062
min    2023-01-01 00:00:00    654.220000  ...    69.080000    0.630000
25%    2023-01-06 20:00:00   1708.980000  ...   240.900000    8.230000
50%    2023-01-12 16:00:00   2590.180000  ...   340.900000   14.820000
75%    2023-01-18 12:00:00   4432.680000  ...   482.570000   26.350000
max    2023-01-24 08:00:00  16876.220000  ...  1499.270000  267.510000
std                    NaN   3227.744681  ...   271.287026   36.563094

[8 rows x 9 columns]
```

```
                      date            co  ...         pm10         nh3
count                  561    561.000000  ...   561.000000  561.000000
mean   2023-01-12 16:00:00   3814.942210  ...   420.988414   26.425062
min    2023-01-01 00:00:00    654.220000  ...    69.080000    0.630000
25%    2023-01-06 20:00:00   1708.980000  ...   240.900000    8.230000
50%    2023-01-12 16:00:00   2590.180000  ...   340.900000   14.820000
75%    2023-01-18 12:00:00   4432.680000  ...   482.570000   26.350000
max    2023-01-24 08:00:00  16876.220000  ...  1499.270000  267.510000
std                    NaN   3227.744681  ...   271.287026   36.563094

[8 rows x 9 columns]
```



INTERACTIVE POLLUTANT TRENDS OVER TIME



CORRELATION BETWEEN POLLUTANTS

7. Interpretation and Analysis:

- The **high concentrations of PM2.5 and PM10** observed throughout January suggest that particulate matter pollution is a chronic problem in Delhi. These pollutants are typically emitted by vehicles, industrial processes, and the burning of waste, which are prevalent in urban environments like Delhi.
- The **spikes in CO** likely correspond to events of high vehicle emissions or industrial activity. These spikes are dangerous as they pose immediate health risks during exposure.
- **NO2 and O3**, although present in lower quantities, are still concerning, especially given their cumulative impact on respiratory health and their role in forming secondary pollutants like ozone at ground level.

6. Conclusions and Recommendations:

Conclusions:

- Delhi experiences **critical air quality issues**, especially concerning particulate matter (PM2.5 and PM10), which consistently exceed safe limits and pose significant health risks.
- The **spikes in CO and NO2** are likely linked to traffic and industrial activities, which worsen during certain periods.
- **Immediate interventions** are necessary to manage pollutant levels and reduce public health risks.

Recommendations:

1. **Introduce Emission Control Measures:**
    - **Reduce vehicular emissions** by promoting electric vehicles and implementing stricter regulations on diesel and petrol vehicles.
    - **Control industrial emissions** through better regulation, cleaner technologies, and stricter enforcement of environmental laws.
2. **Improve Public Transportation:**
    - Promote the use of public transport, carpooling, and non-polluting alternatives (e.g., bicycles) to reduce the number of vehicles on the road.
3. **Monitor and Control Waste Burning:**
    - Implement stricter bans and monitoring systems to reduce the burning of solid waste and agricultural residues, which contribute heavily to particulate pollution.
4. **Raise Public Awareness:**
    - Increase awareness campaigns on the impact of pollution on health and how individuals can reduce their personal contribution to air pollution.
5. **Enhance Green Spaces:**
    - Increase the number of green spaces and promote urban forestry to help reduce the concentration of pollutants, especially in densely populated areas.